

ELECTROMYOGRAPHY & NERVE CONDUCTION VELOCITY

EMG & NCV



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Objective:

- Assess and record electrical activity of the neuromuscular system
- Measure muscle activity
- Evaluate nerve conduction
- Detect abnormalities of:
 - Peripheral nerves
 - Neuromuscular junction
 - Skeletal muscle

Electromyography (EMG) and Nerve conduction velocity (NCV)



- ❑ The human body acts as an **inhomogeneous volume conductor**, allowing ionic currents to generate **measurable electrical potentials on the body surface**.
- ❑ **Active muscles produce electrical potentials**, which are generally **proportional to the level of muscle activity**.
- ❑ **Electromyography (EMG)** is the technique used to **record and analyze the electrical activity of muscle tissue**.

❑ Muscle

- ❑ Composed of **several motor units**

❑ Motor unit

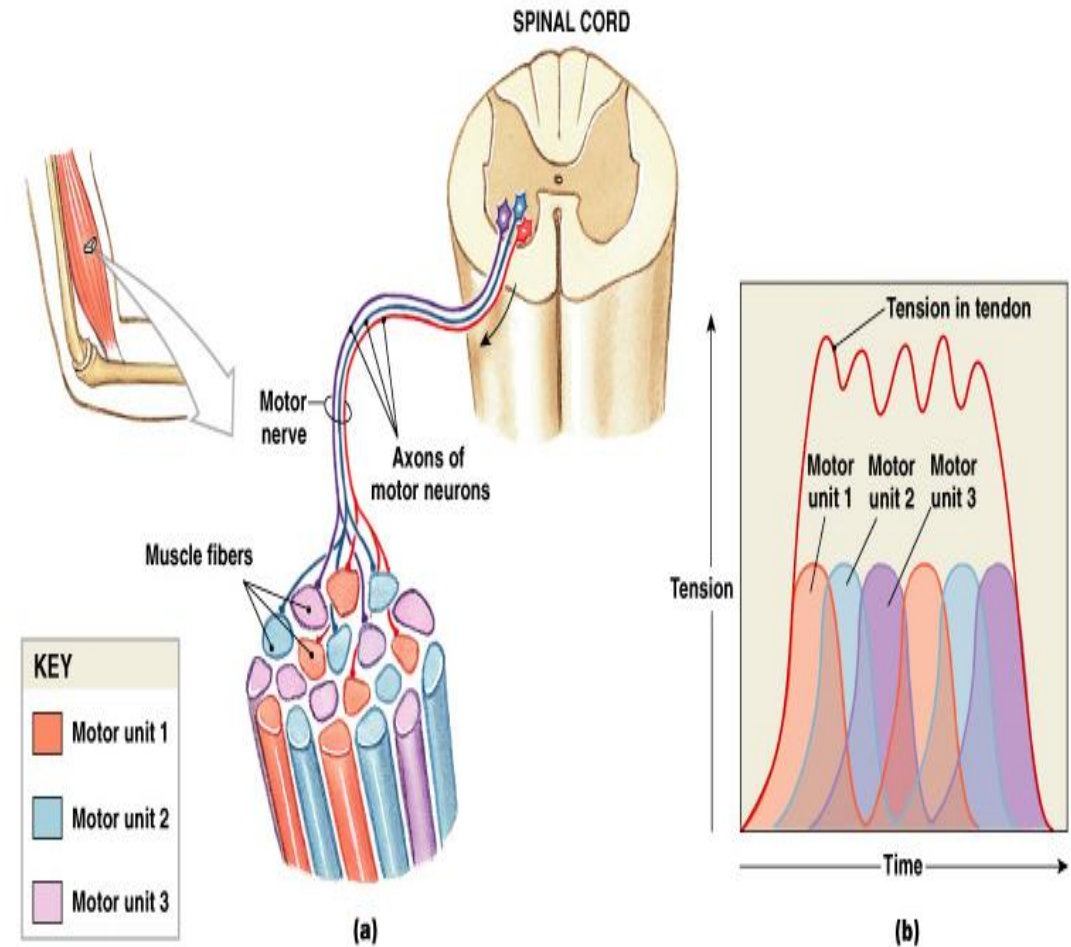
- ❑ Consists of a **single motor neuron** and the **muscle fibers** it innervates

❑ Motor neuron

- ❑ Originates in the **brainstem or spinal cord**
- ❑ Innervates approximately **4–2000 muscle fibers**

❑ Neuromuscular junction

- ❑ The **synaptic junction** between the motor neuron and muscle fibers
- ❑ Also called the **motor end plate**



- Electromyography (EMG) is a valuable tool for detecting and diagnosing abnormalities, not only in the muscles themselves but also in the broader motor system.
 - EMG involves the recording and interpretation of muscle action potentials, providing insights into muscle function and the integrity of the nerves that control them.

ELECTRICAL CHARACTERISTICS



- The electrical source is the muscle membrane potential of about -70mV.
 - Measured EMG potentials range between 50 μ V up to 30 mV, depending on the muscle under observation.
- Typical repetition rate of muscle unit firing is about 7-20 Hz (size of the muscle).
- Damage to motor units can be expected at ranges between 450 and 780 mV

Nerve conduction studies

- Peripheral nerves stimulated with controlled electrical stimulus.
 - Generally useful for distal/extremity nerves only e.g.(median, ulnar, radial) in upper limb,- (tibial, Peroneal, sural) in lower limbs
- Current depolarizes nerve membrane & initiates action potential in axon
- Action potential travels along nerve
 - Speed on conduction depends on integrity of insulating myelin membrane to allow salutatory conduction
 - Conduction along non-myelinated fibers much slower

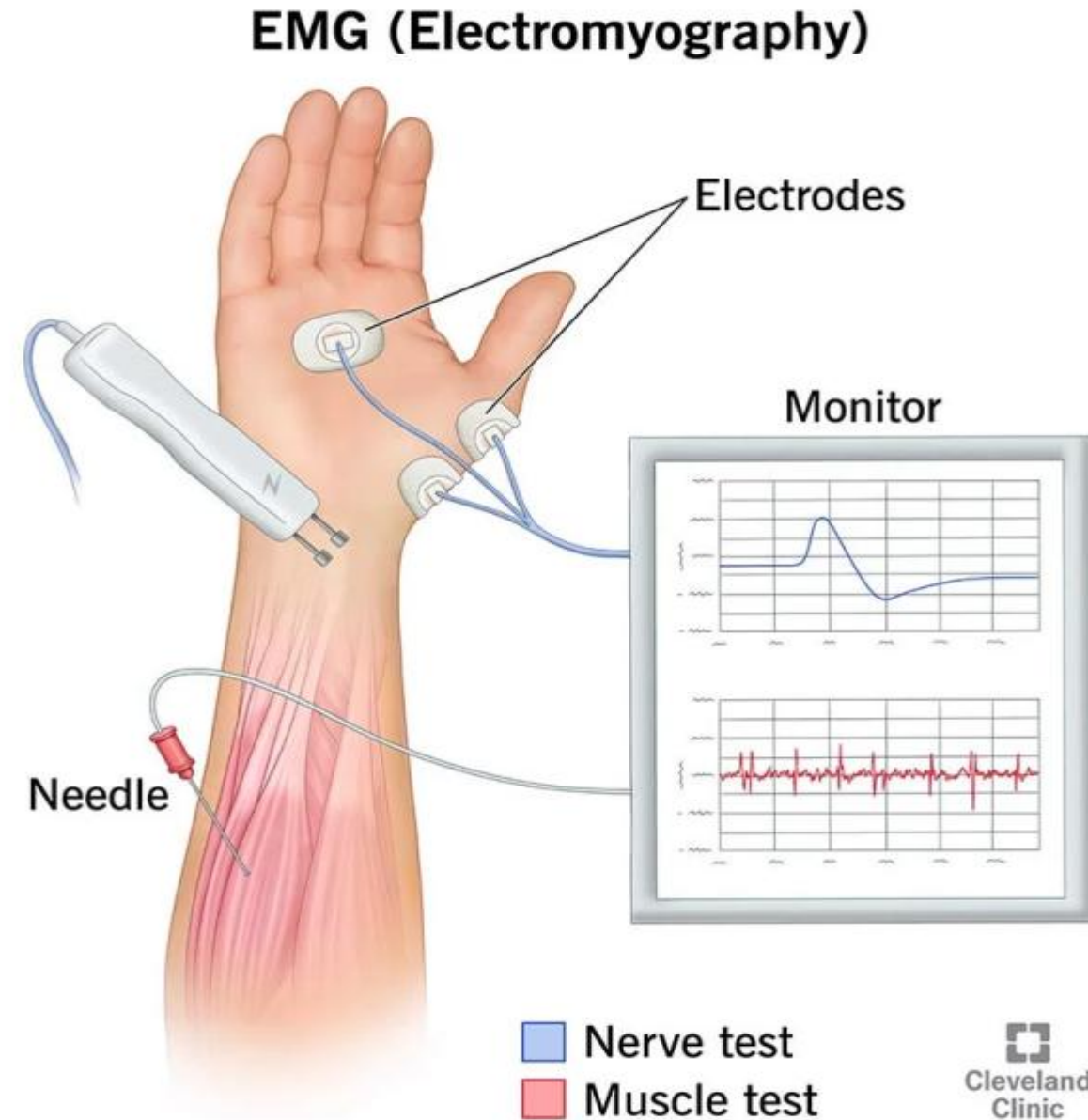
- There are 2 main application of EMG

- Observation of electric potential fluctuation as a function of time and in response to a muscle contraction
- Measurement of signal propagation along nerve fibers after stimulation

- When pathologic conditions arise in the motor system, whether in the
 - spinal cord,
 - the motor neurons,
 - the muscle, or
 - the neuromuscular junctions,
- the characteristics of the electrical signals in the muscle change.

Recording Methodology

- Electrical potential difference measured between two points → bipolar electrode configuration used
- Bipolar Electrode Types
 - Needle
 - Surface
- Electrode Placement
 - Overlying the muscle of interest in the direction of predominant fiber direction
 - Subject is **GROUNDED** by placing an electrode in an inactive region of body



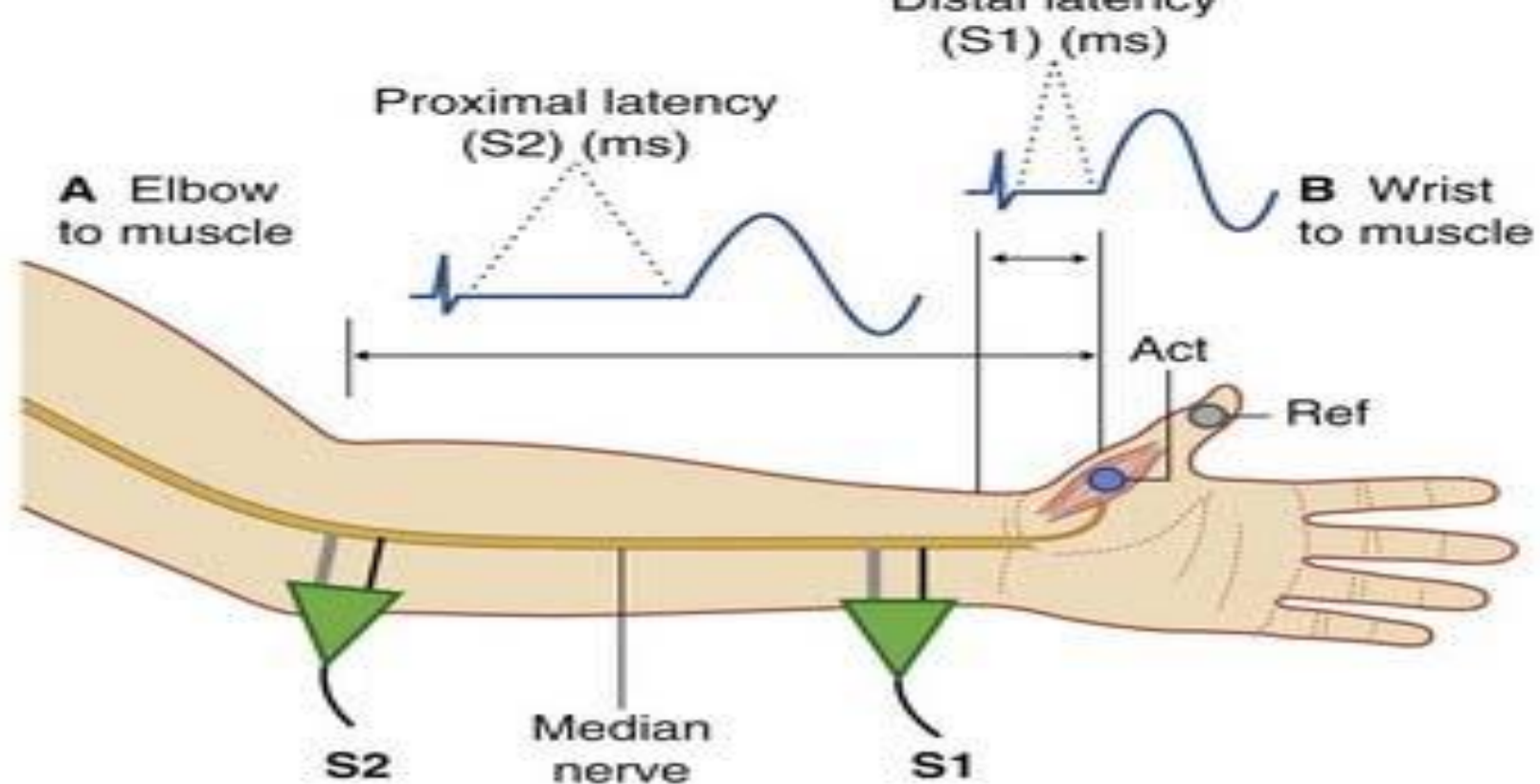
- More forceful contractions lead to greater action potential activity.
- A typical stimulating pulse may have an amplitude of 100V and last 0.1 to 0.5ms.
 - With electrical stimulation the stimulation time is well defined and
 - all the muscle fibres fire at nearly the same time.
- The action potential appears after a latency period time.
- A voluntary contraction is usually spread over about 100 ms because all the motor units do not fire at the same time,
- Also, each motor unit produce several action potentials depending upon the signals send from the CNS

Analyzing the EMG Signal

- Amplitude & Frequency
- More motor unit → more amplitude, more spikes and more turns in signal
- Change in firing rate → change in frequency content of EMG
- Change in muscle fiber type → change in action potential velocity, change in EMG frequency.
- EMG is spatial and temporal summation of action potentials

Nerve Conduction Study

- Responses are recorded
 - Motor studies: Compound motor action potential (CMAP)
 - Sensory studies: Sensory nerve action potential (SNAP)
 - H-reflex
- Assess large myelinated fibers only
 - Not useful for small fiber disorders
- Assess for:
 - axonal processes (axon loss, Wallerian degeneration)
 - Demyelinating processes (slowing, conduction block)



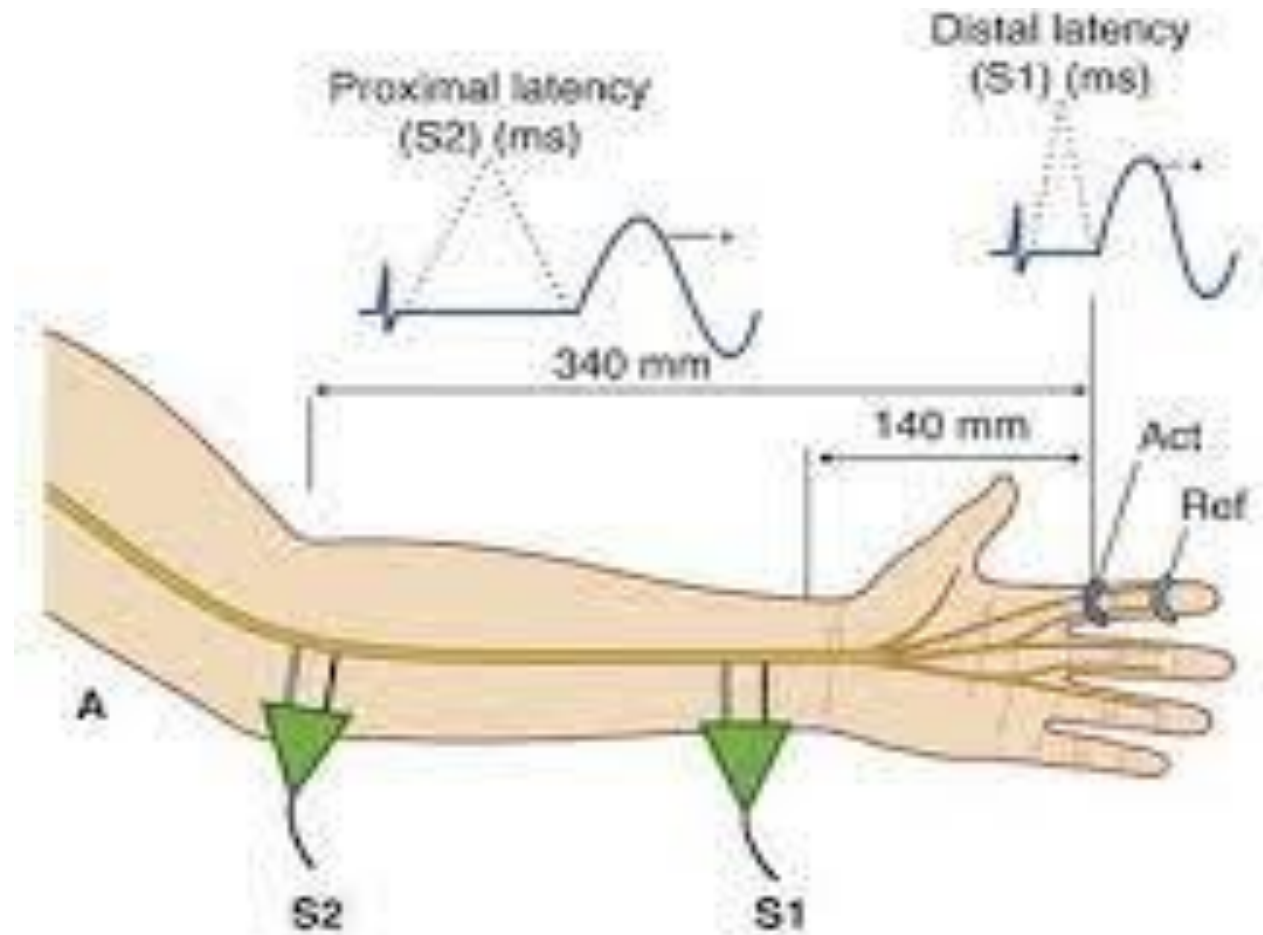
MNCV = Motor nerve conduction velocity

MNCV case example:

$$\text{MNCV} = \frac{\text{Distance (A-B)}}{\text{Time (A-B)}} = \frac{340 - 40 \text{ mm}}{10 - 4 \text{ ms}} = \frac{300 \text{ mm}}{6 \text{ ms}}$$

MNCV = 50 m/s for median nerve segment S1-S2

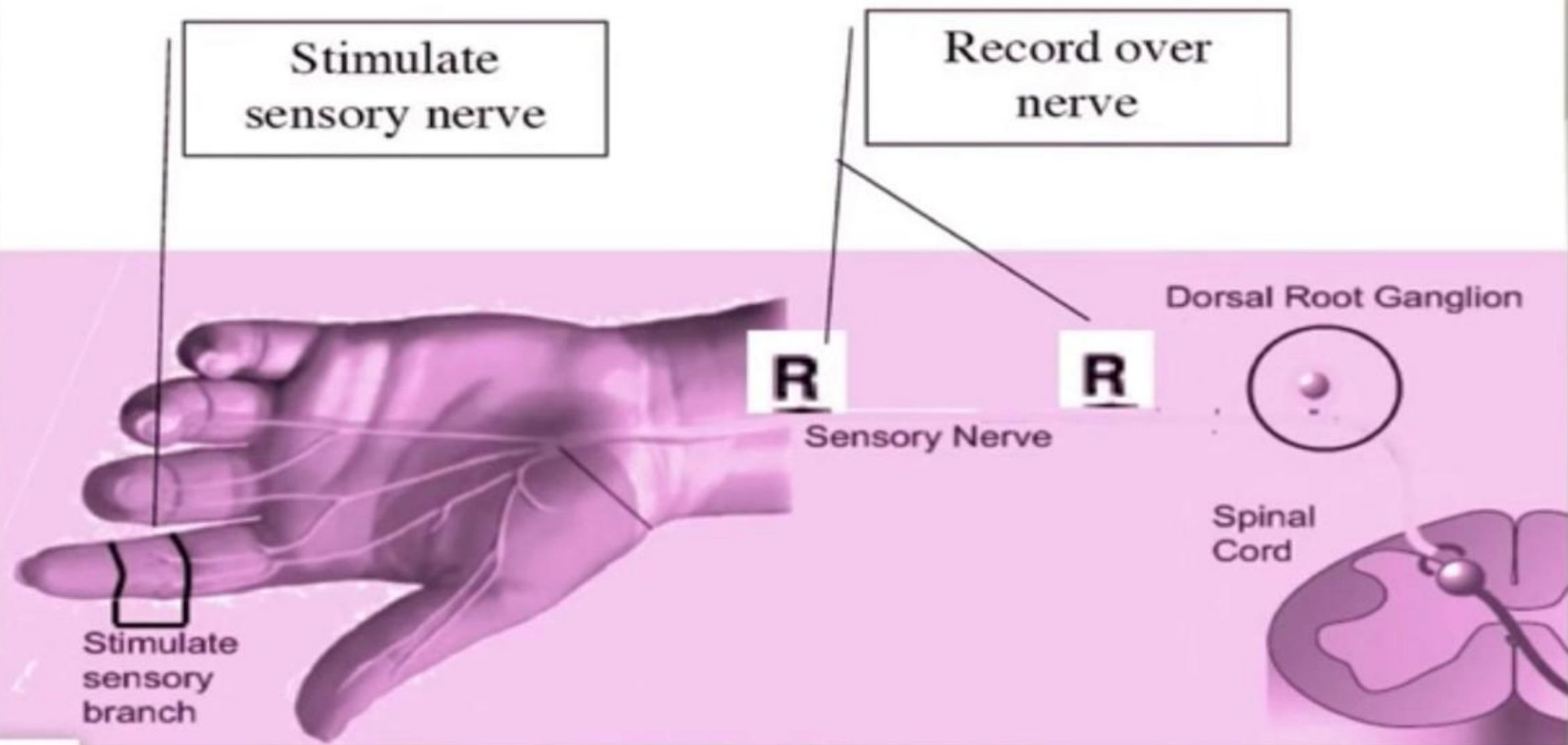
Motor Nerve Conduction Study



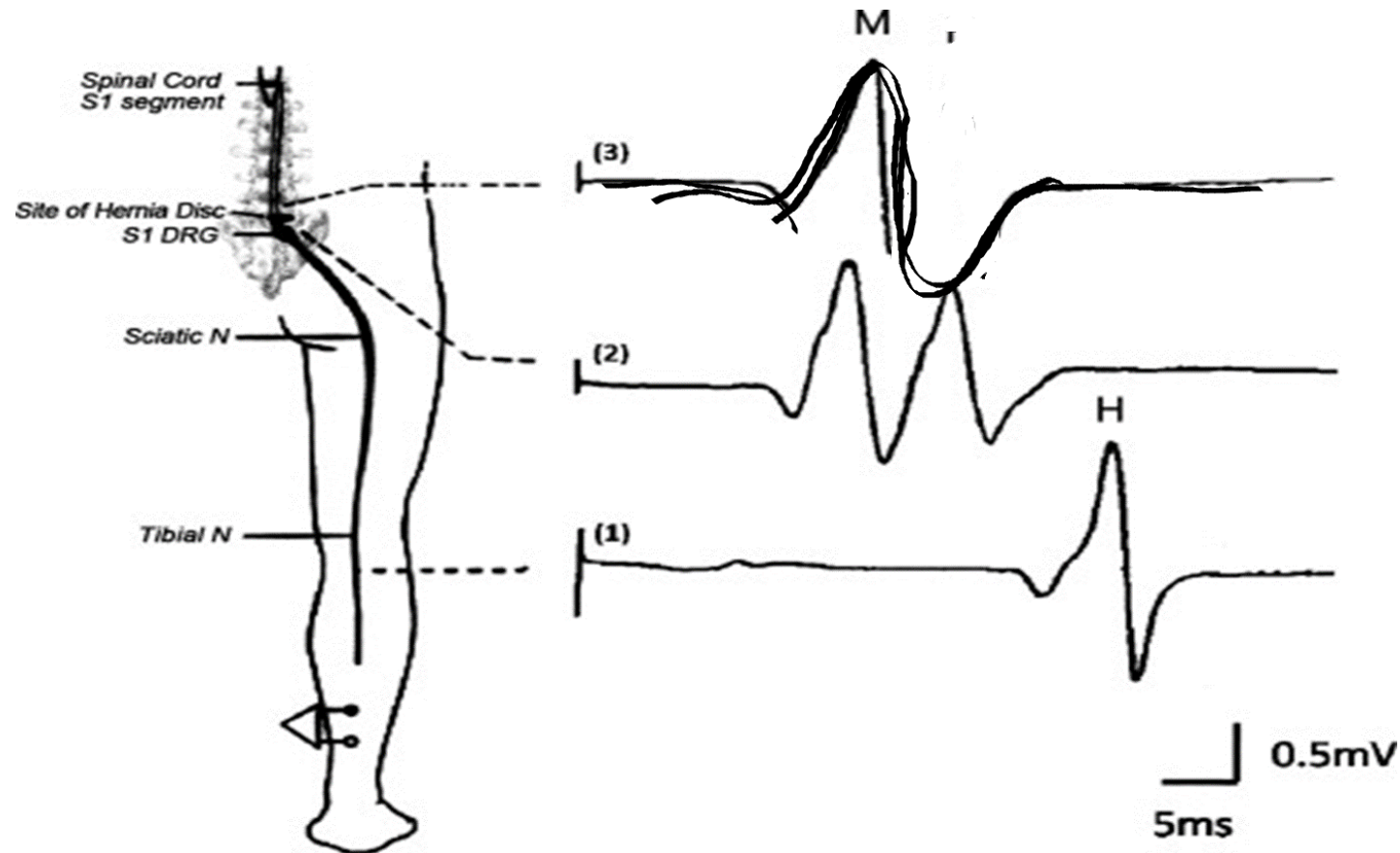
Case example:

$$\text{SNCV} = \frac{340 - 140 \text{ mm}}{5.8 - 2.5 \text{ ms}} = \frac{200 \text{ mm}}{3.3 \text{ ms}} = 61 \text{ m/s}$$

Sensory nerve



Sensory reflex system Study



A 62-year-old man has **6 months of numbness and mild weakness** with **absent reflexes**. Nerve conduction studies show

- **markedly slowed conduction velocity,**
- **prolonged distal and F-wave latencies,**
- **and conduction block,** with
- **relatively preserved amplitudes.**

These findings are most consistent with which **underlying process**?

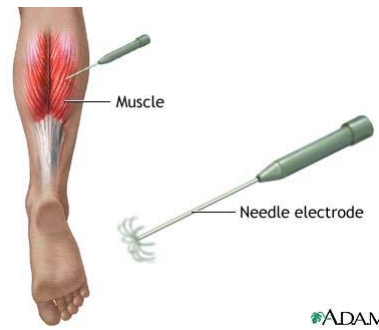
- A. Primary muscle fiber degeneration
- B. Acute axonal degeneration
- C. Chronic demyelination of peripheral nerves
- D. Anterior horn cell degeneration
- E. Neuromuscular junction failure

Correct Answer:

☐ **C. Chronic demyelination of peripheral nerves**

Needle Electromyography (Needle EMG)

- **Procedure:** A needle electrode is inserted directly into the muscle.
- **Electrode type:** Disposable, single-use needles; commonly **concentric** electrodes, which combine recording and reference within the same electrode.
- **Muscle selection:** Multiple muscles can be examined. The choice of muscles depends on the **clinical question**.
- **Patient experience:** Discomfort is generally **mild**.
- **Muscle activity studied:**
 - **At rest:** Muscle should be silent. Spontaneous activity may indicate **nerve or muscle abnormalities**.
 - **During contraction:** Muscle is studied at different levels of **sustained, voluntary contraction**.
- **Interpretation:** The **electrical activity, shape, and pattern of responses** help differentiate **nerve disorders** from **muscle diseases**.



Needle Electromyography (EMG) – Parameters Evaluated

1. Motor Unit Configuration

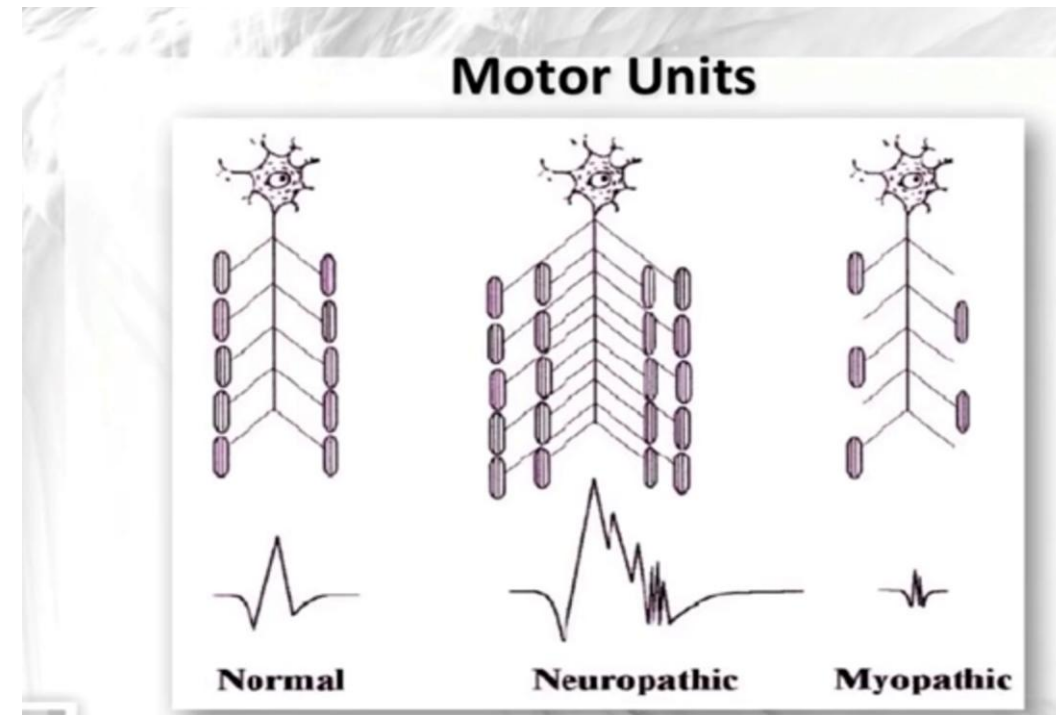
- Muscle is **volitionally activated** at different force levels.
- **Single motor units** are assessed, recording the **motor unit action potential (MUAP)**, which reflects the activity of all muscle fibers innervated by a single motor axon.

2. Parameters of MUAP

- **Amplitude**
- **Duration**
- **Morphology**

3. Patterns Observed

- **Normal:** Normal amplitude, duration, and shape
- **Neuropathic:** High amplitude, long duration, often **polyphasic** (multiple phases)
- **Myopathic:** Low amplitude, short duration, often **polyphasic**



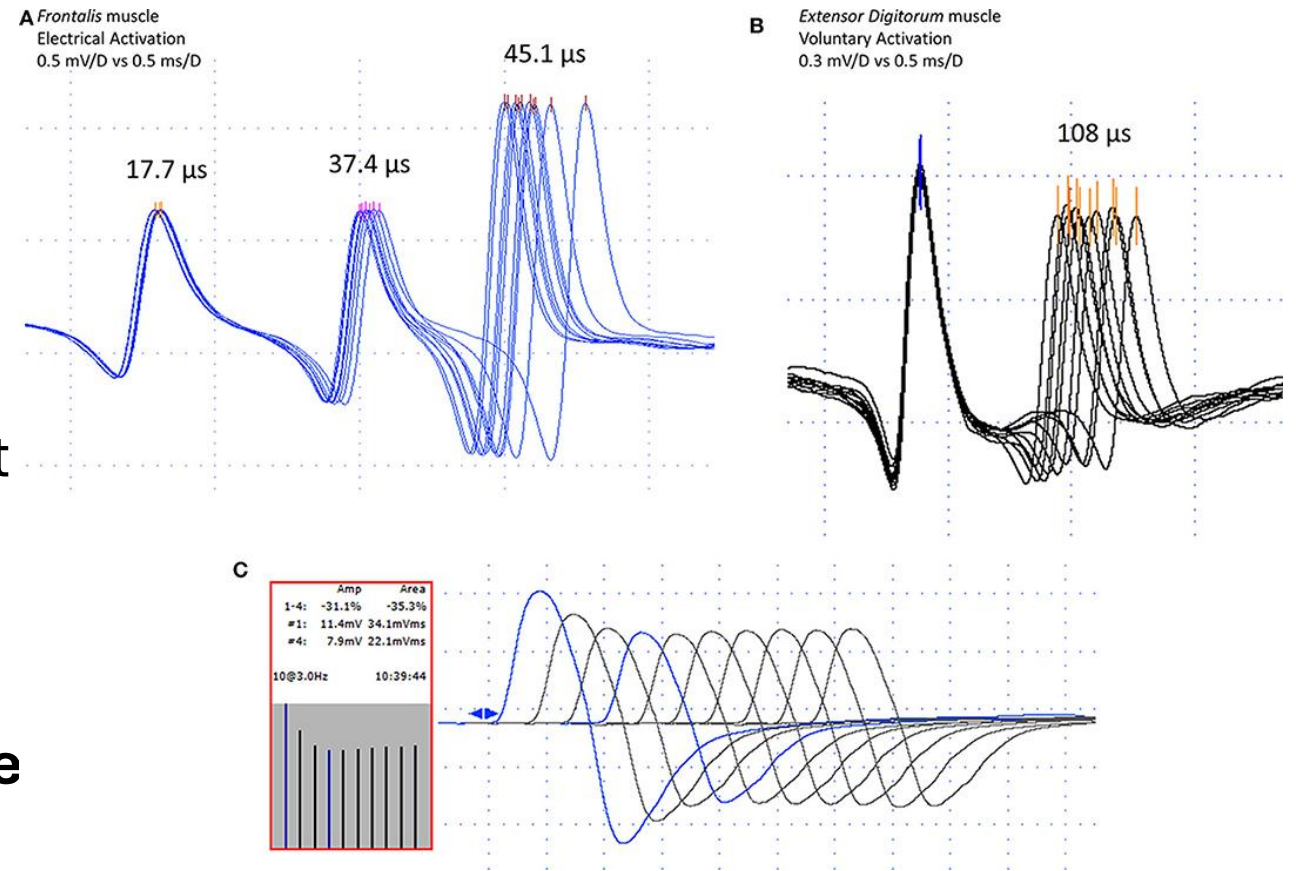
Motor Unit Recruitment and Interference Pattern

1. Normal Recruitment

- Graded recruitment: **small MUAPs appear first**, followed by **larger MUAPs** as force increases.
- **Full interference pattern** is achieved at maximal voluntary contraction.

2. Reduced Recruitment

- Characterized by **large MUAPs at high firing rates** but **incomplete interference pattern**.
- Typically seen in **neuropathic conditions**.

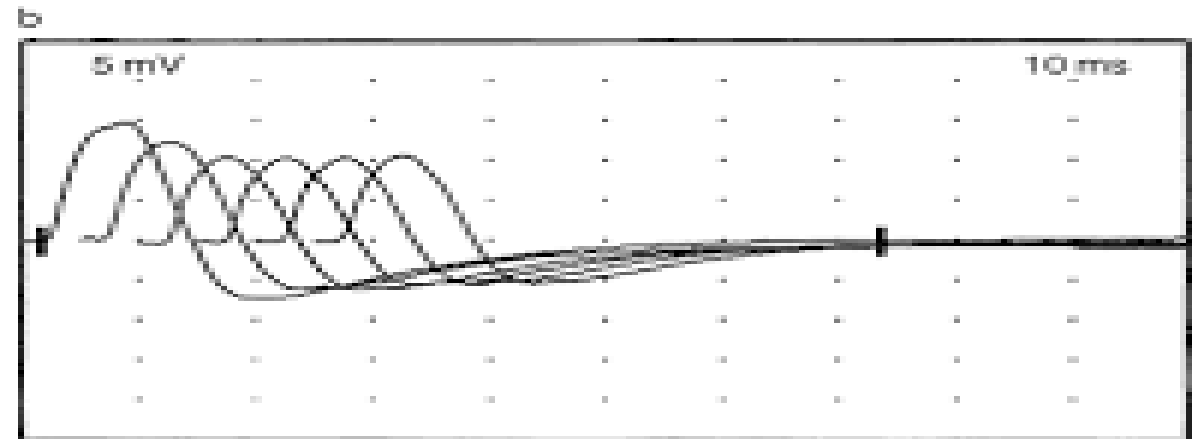
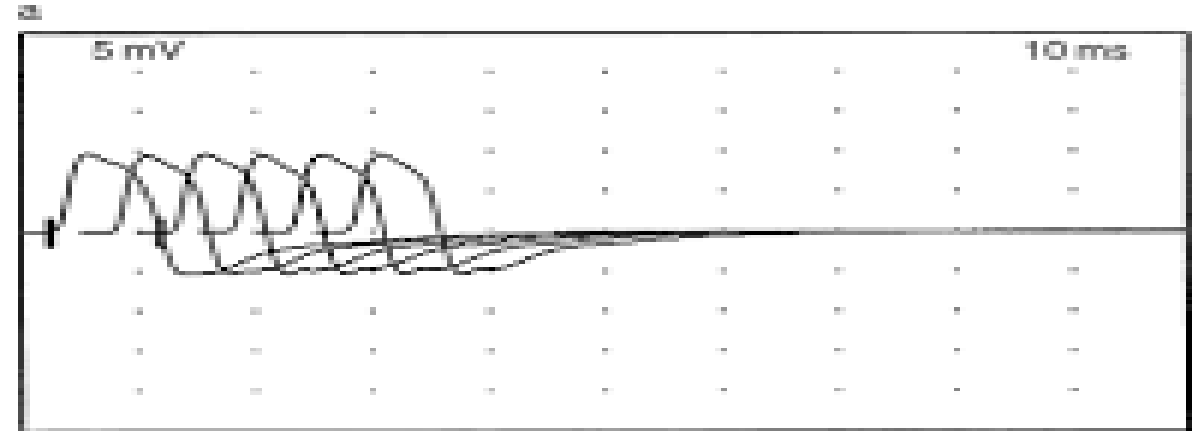


3. Increased (Early) Recruitment

- **Many small MUAPs** are recruited early during minimal contraction.
- Commonly observed in **myopathy**.

4. Additional Notes

- Recruitment patterns are **dependent on patient cooperation and effort**.
- Analysis helps **differentiate neuropathic from myopathic disorders**.



Myasthenia Gravis

A 35-year-old man presents with progressive proximal muscle weakness. EMG shows:

- Low-amplitude, short-duration motor unit potentials
- Polyphasic potentials
- Early recruitment
- No fibrillation potentials

These findings are most consistent with:

- A. Acute denervation
- B. Chronic denervation
- C. Neuromuscular junction disorder
- D. Myopathy
- E. Demyelinating neuropathy

 **Answer: D. Myopathy**

Explanation: Reduced amplitude, short duration, and early recruitment without fibrillation are classic myopathic features.

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